Cross-Platform Arbitrage in Cryptocurrency Prediction Markets: An Empirical Analysis

Detecting Latency Arbitrage, Cross-Platform Trade Synchronization, and Coordinated Wallet Networks Across Polymarket and Kalshi

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Abstract

We present an empirical analysis of cross-platform arbitrage in cryptocurrency prediction markets, documenting two distinct arbitrage channels: **Binance → Polymarket latency arbitrage** on 5-minute BTC markets, and **Polymarket ↔ Kalshi temporal synchronization** on 15-minute BTC markets. Our dataset spans two periods—a primary analysis (Feb 19–25, 2026) and an out-of-sample validation week (Feb 25–Mar 2, 2026)—covering approximately **10.5 million trades** in total: Polymarket 5-min (2,182,329 trades, 26,475 wallets, 642 windows), Polymarket 15-min (4,751,627 trades across both weeks), and Kalshi KXBTC15M (3,574,785 trades across both weeks).

On 5-minute markets, we identify 29 wallets achieving >98% win rates by trading in the final 15 seconds of each window, using Binance BTC price as a free information feed. These wallets show 97–100% win rates when Binance moves >$80, but ∼50% (random) on small moves—their entire alpha comes from real-time exchange data.

On 15-minute markets, we perform second-by-second cross-platform trade matching and discover **161,577 exact same-second trade pairs** between Polymarket and Kalshi in week 1, replicated with **148,142 same-second pairs** in the out-of-sample week 2. We find **1,368 opposite-side amount-matched trades** in week 1 and introduce a tiered matching methodology in week 2 yielding **357,724 opposite-side matches** across four tolerance tiers. We identify 1,810 Polymarket wallets with >58% Kalshi outcome alignment in week 1 and 1,261 in week 2, with wallets persisting across both periods at 100% alignment. Despite 6.9% average price spreads between platforms, the 5.7–6.2% oracle disagreement rate (Chainlink vs. CF Benchmarks BRTI) makes cross-platform hedging non-risk-free. Crucially, the out-of-sample week reveals a **declining PnL trend** (−$31/window/day), suggesting the arbitrage opportunity is being competed away even as the structural mechanism persists.

1 Overview

We analyze **approximately 10.5 million trades** across Polymarket, Kalshi, and Binance over two periods (Feb 19–25 and Feb 25–Mar 2, 2026) to document two arbitrage channels:

1. **Binance → Polymarket (5-min):** 29 wallets achieve >98% win rates by watching Binance BTC price and trading Polymarket in the final seconds before resolution.
2. **Polymarket ↔ Kalshi (15-min):** 161,577 same-second trade pairs and 1,368 opposite-side amount-matched trades across 672 matched windows—direct evidence of automated cross-platform arbitrage.

1.1 Platform Structure

Table 1: BTC binary market platforms

Feature	Polymarket	Kalshi	Binance
Shortest BTC interval	5 min	15 min	N/A (spot)
Settlement oracle	Chainlink	CF Benchmarks BRTI	N/A
Maker fee	0% + rebate	Free	0.1%
On-chain traceability	Full (Polygon)	None	None
Trade anonymity	Pseudonymous	KYC required	KYC required

Polymarket's 5-min BTC markets are NegRisk binary options on Polygon (Chainlink oracle, CLOB with EIP-712 orders, 1 UP + 1 DOWN = 1 USDC). Kalshi's shortest BTC market is 15-min (`KXBTC15M`), settled by CF Benchmarks BRTI. The different oracles create a 5.7% resolution disagreement rate—critical for cross-platform arb risk.

1.2 Research Questions

1. **Latency arbitrage:** Do wallets systematically exploit the lag between BTC price movements on centralized exchanges and Polymarket order book updates?
2. **Cross-platform hedging:** Can we identify wallets whose on-chain behavior is consistent with simultaneous positions on Polymarket and other platforms (Kalshi, Binance futures)?
3. **Market-maker networks:** Do coordinated wallet clusters exist where one entity provides liquidity while another captures it?
4. **Information asymmetry:** Does win rate vary systematically with Binance move magnitude, suggesting exchange-informed trading?

1.3 Contributions

- **Binance → Polymarket (5-min):** Identification of 1,647 latency arbitrage candidates from 26,475 wallets, including 29 with >98% win rates trading in the final seconds using real-time Binance BTC prices.
- **Polymarket ↔ Kalshi (15-min):** Cross-platform trade synchronization analysis finding 161,577 same-second trade pairs and 1,368 opposite-side amount-matched arb signals—the direct signature of automated cross-platform arbitrage.
- Identification of 1,810 Polymarket wallets with anomalous Kalshi outcome alignment (>58%), including wallets with 100% win rate and full Kalshi alignment across hundreds of trades.
- Discovery of a coordinated market-making network where a single counterparty takes the opposite side of latency bots in 99.8% of shared markets.
- Quantification of the 5.7% oracle disagreement rate between Chainlink and CF Benchmarks BRTI that makes cross-platform hedging non-risk-free.
- A six-hypothesis testing framework for detecting arbitrage behavior in prediction market trade data.
- Per-strategy P&L decomposition accounting for fees, gas, and slippage; temporal stationarity analysis showing whether arb opportunities persist or decay; and a loss distribution quantifying who funds arbitrage profits (market makers, retail, other bots).
- Out-of-sample validation on a second week of data (∼4.1M trades, Feb 25–Mar 2) confirming structural persistence of cross-platform synchronization while revealing declining per-trade profitability (−$31/window/day trend).

2 Data and Methodology

2.1 Dataset

Our dataset spans two periods: a primary analysis week (February 19–25, 2026) and an out-of-sample validation week (February 25–March 2, 2026), comprising four data sources:

Table 2: Dataset summary

Source	**Metric**	**Value**
Polymarket 5-min (Week 1)	Trades	2,182,329
	Unique wallets	26,475
	Market windows	642
	Boundary records (w/ Binance)	929
Polymarket 15-min (Week 1)	Trades	2,536,367
	Markets found	672
	Resolutions (Up / Down)	353 / 319
	Avg trades per market	3,774
Kalshi KXBTC15M (Week 1)	Trades	1,680,127
	Markets	673 (665 with trades)
	Total volume (contracts)	88,050,035
	Avg volume per market	130,832
	Max volume (single market)	426,659
Polymarket 15-min (Week 2)	Trades	2,215,260
	Markets found	576
	Unique wallets	19,138
Kalshi KXBTC15M (Week 2)	Trades	1,894,658
	Markets	633
Week 1 total trades		**6,398,823**
Week 2 total trades		**4,109,918**
Grand total trades		**∼10.5M**
Date range		Feb 19–Mar 2, 2026
Matched cross-platform windows (W1 / W2)		672 / 564

Polymarket. Trade data was collected from the Gamma API and Data API, with per-trade fields: `proxyWallet`, `side` (BUY/SELL), `asset` (token ID), `size`, `price`, `timestamp`, and `transactionHash`.

Kalshi. Trade data from the public REST API[1] per ticker: `created_time`, `yes_price`, `no_price`, `taker_side`, `count`, `trade_id`. Kalshi trades are anonymous (no user identifiers), preventing direct wallet-to-account matching.

Boundaries. Chainlink and Binance start/end prices per 5-minute window, enabling ground-truth direction labels.

[1] `/trade-api/v2/markets/trades`

2.2 Hypothesis Testing Framework

We test six hypotheses for arbitrage behavior, each with a distinct detection signature:

Table 3: Hypothesis testing framework

H#	Name	Detection Signature
H1	Latency Arbitrage	>85% WR in last 30s, >90% WR in last 15s, >10 late trades
H2	Sell-Cluster Arb	>40% sell ratio, >60% sell WR, systematic loser-token selling
H3	Binance Correlation	WR on large BN moves (>$80) exceeds WR on small moves (<$30) by >10pp
H4	Cross-Platform Sig	Round sizes, consistent timing (std < 60s), >65% WR, >50 markets
H5	Market Making	Both-side trading in >10% of windows, >3 both-side windows
H6	Divergence Exploit	Higher WR on high CL–BN divergence (>$20) windows

For each wallet with ≥20 trades, we compute a composite score per hypothesis and flag wallets exceeding threshold criteria. Wallets appearing across multiple hypotheses receive the strongest arbitrage classification.

2.3 Win Rate Computation

For each trade, we determine the *bet direction*:

- **BUY** side → bet direction = token direction (buying UP means betting UP)
- **SELL** side → bet direction = opposite of token direction (selling UP means betting DOWN)

A trade *wins* if its bet direction matches the market resolution.

3 Results

3.1 H1: Latency Arbitrage (Exchange → Polymarket)

Hypothesis: Some wallets exploit the lag between BTC price movements on centralized exchanges and Polymarket's order book adjustment, trading after the direction is known but before the market resolves.

Result: Confirmed. We identify 1,647 latency arb candidates, of which 29 achieve >98% win rates.

Table 4: Top latency arbitrage wallets (sorted by arb score)

Pseudonym	Trades	WR	Last-15s	L15 WR	Avg t	Buy%	Vol
Ambitious-Birch	132	100.0%	123	100.0%	293s	100%	$261
Plaintive-Forgery	304	99.0%	116	100.0%	278s	100%	$3,000
Nippy-Styling	141	98.6%	141	98.6%	299s	100%	$1,335
Swift-Consist	465	98.7%	126	98.4%	247s	100%	$24,313
Angry-Melatonin	361	99.2%	—	—	248s	100%	$2,799
Impeccable-Venison	1,571	90.9%	1,064	96.5%	238s	100%	$1,751
Nocturnal-Law	2,331	98.2%	—	—	277s	100%	$24,435
Long-Lark	220	95.5%	129	99.2%	285s	100%	$9,622

Key evidence: Nippy-Styling trades at an average of **299.2 seconds** into each 300-second window—literally 0.8 seconds before resolution—with 98.6% win rate across 141 trades. All 141 trades occur in the 285–300s bucket. This timing is only possible if the wallet operator observes the BTC price on centralized exchanges (where BTC trades continuously) and submits a Polymarket order in the final second, after the direction is already determined by the exchange price.

Ambitious-Birch achieves **100% win rate across 132 trades**, with 123 of those in the last 15 seconds (average entry: 293s). A binomial test yields $p < 10^{-30}$ for this being achieved by chance, given the ∼50% base rate of BTC 5-minute direction.

3.2 H2: Sell-Cluster Arbitrage

Hypothesis: Wallets systematically sell losing tokens (NegRisk minting) rather than buying winners, either for better execution prices or as part of a cross-platform hedge.

Result: Confirmed with nuance. We find 2,054 sell-heavy wallets (>40% sells), of which 390 achieve >60% sell win rate.

Table 5: Top sell-cluster wallets

Pseudonym	Trades	Sells	Sell%	Sell WR	Buy WR	Sell Vol
Capital-Potty	26,816	14,360	54%	79.2%	22.9%	$51,123
Spry-Incident	15,366	8,436	55%	79.0%	23.4%	$40,512
Guilty-Expertise	3,900	2,692	69%	69.1%	47.6%	$17,439
Numb-Biopsy	68	68	100%	82.4%	—	$301

Capital-Potty is notable: 79.2% sell win rate but only 22.9% buy win rate. This inverted pattern suggests systematic sell-the-loser strategy. Their sells are profitable; their buys are the complementary token they don't want (or they're providing the buy side as a market maker).

Numb-Biopsy is a **pure sell-only wallet** (100% sells, 0% buys) with 82.4% win rate—a textbook sell-the-loser arbitrage pattern using NegRisk token minting.

3.3 H3: Binance Price-Move Correlation

Hypothesis: Wallets trade preferentially when BTC has already moved significantly on Binance, using exchange momentum as a directional signal.

Result: Strongly confirmed. We find 1,982 Binance-correlated wallets.

Table 6: Win rate by Binance move magnitude (selected wallets)

Pseudonym	Trades	WR	|BN| <30	WR	|BN| >80	WR	Δ
(anonymous)	283	81.3%	20	40.0%	192	**97.9%**	+57.9pp
Slim-Recovery	110	87.3%	35	60.0%	39	**100.0%**	+40.0pp
Dead-Grandpa	88	87.5%	23	78.3%	23	**100.0%**	+21.7pp
Alive-Recall	665	70.5%	261	49.4%	199	**81.9%**	+32.5pp
Nifty-Union	117	78.6%	42	45.2%	32	**100.0%**	+54.8pp

The anonymous wallet (0xe...) is the clearest signal: **40% win rate when BTC moves less than $30** (essentially random) but **97.9% win rate when BTC moves more than $80** (near-certain). This wallet has zero edge when the exchange signal is ambiguous—their entire alpha comes from Binance move magnitude. Nifty-Union shows the same pattern: 45.2% (random) on small moves, 100% on large moves.

This provides strong evidence that these wallets are making trading decisions *conditional* on real-time centralized exchange price data.

3.4 H4: Cross-Platform Arbitrage Signatures

Hypothesis: Wallets operating cross-platform arbitrage show distinctive patterns: round trade sizes (automated), consistent timing (scheduled), and high market coverage (systematic).

Result: 1,860 candidates identified. The strongest signal comes from a cluster of wallets with identical behavioral profiles:

Table 7: Suspected bot cluster with identical behavior profiles

Pseudonym	Trades	WR	Avg Size	Round%	σ_t	Avg t	Sell%
Turbulent-Solidarity	295	80.0%	35.5	84%	25s	67s	100%
Grotesque-Satellite	320	81.6%	55.4	81%	26s	64s	100%
Black-And-White-Laborer	284	81.7%	34.5	83%	26s	64s	100%
Angry-Livestock	295	77.6%	62.5	82%	27s	62s	100%
Extra-Small-Coffee	298	78.9%	57.2	77%	25s	63s	100%
Automatic-Fairy	295	80.0%	36.9	84%	24s	64s	100%

These six wallets share nearly identical statistics: all 100% sell-only, 77–82% win rate, timing standard deviation of 24–27 seconds, average entry at 62–67 seconds, and 77–84% round trade sizes. The identical behavioral fingerprint strongly suggests these are **the same operator running multiple wallets**—likely a sell-the-loser minting strategy that distributes capital across wallets to avoid position limits or detection.

3.5 H5: Market Makers and Two-Sided Traders

Result: 3,716 two-sided traders identified.

Table 8: Top market makers by volume

Pseudonym	Windows	Both-Side	Trades	Buy Vol	Sell Vol
Canine-Commandment	636	636 (100%)	38,170	$1,132,047	$0
Square-Guy	582	576 (99%)	23,197	$924,890	$0
Poised-Twist	625	625 (100%)	34,778	$557,149	$220,311
Novel-Desktop	492	490 (100%)	6,757	$445,041	$7,080
Lumbering-Stepdaughter	522	522 (100%)	17,761	$166,908	$0

Canine-Commandment trades *both* UP and DOWN in every single window (636/636 = 100%), with $1.13M in buy volume. "Buy Vol" and "Sell Vol" refer to the CLOB order side (BUY or SELL), not the token direction. Wallets with $0 sell volume execute exclusively via BUY orders on *both* the UP and DOWN tokens—buying UP is a bullish bet, buying DOWN is a bearish bet. This is textbook automated market making: providing liquidity on both sides of the book in every window via limit buy orders.

3.6 H6: Oracle Divergence Exploitation

Result: 829 divergence exploiters identified.

The Binance–Chainlink divergence at boundary shows high variance: mean $195.65, median $7.24, maximum $2,925.77. Windows with >$20 divergence represent 23.6% of all windows.

Several wallets show extreme win rate differentials: one anonymous wallet achieves **95.2% WR on high-divergence windows vs. 1.7% on low-divergence windows**—a 93.5 percentage point gap. This wallet appears to actively *avoid* trading when the oracles agree and aggressively trades when they disagree, exploiting the predictability of the resolution oracle (Chainlink) when it diverges from Binance.

4 Cross-Hypothesis Analysis

4.1 Multi-Hypothesis Wallets

We cross-reference all six hypotheses to identify wallets appearing in multiple categories—the strongest arbitrage signals.

Table 9: Wallets matching ≥5 hypotheses

Wallet	Pseudonym	Hypotheses	Key Signature
0x35c1...	(anonymous)	6/6	87% sell WR, 96% WR large BN
0xe20e...	(anonymous)	6/6	100% sell WR, all 6 hypotheses
0xb689...	Numb-Biopsy	5/6	94% WR large BN, pure sell
0x4b52...	Meek-Plum	5/6	96% WR large BN, 90% overall
0x0d4a...	Whirlwind-Witchhunt	5/6	100% WR large BN, $87 avg

Wallet 0x35c1... triggers **all six hypotheses**: it uses latency timing, sells the loser, performs better when Binance moves are large, shows bot-like patterns, trades both sides, and exploits oracle divergence. This is our strongest arbitrage candidate.

4.2 Coordinated Network Discovery

On-chain interaction analysis reveals a **coordinated trading network** centered on Capital-Potty (0x..., 26,816 trades, $94K volume):

Table 10: Capital-Potty interaction network

Counterparty	Interactions	Opposite-Side	Shared Mkts	Opp%
Impeccable-Venison	4,149	3,774	203	91.0%
Suspect #1	2,349	1,285	20	54.7%
Meek-Plum	1,266	1,239	85	97.9%
Plaintive-Forgery	558	557	195	99.8%
Ambitious-Birch	178	178	108	100.0%

The interaction between Plaintive-Forgery and Capital-Potty is 99.8% opposite-side across 195 shared markets. Ambitious-Birch shows 100% opposite-side across 108 markets. This means **every single time** these latency bots trade, Capital-Potty takes the other side.

Two interpretations are possible:

1. Capital-Potty is an independent market maker, and the latency bots are its natural counterparty (the bots are takers, Capital-Potty is the maker).
2. Capital-Potty and the latency bots are *the same operator*—using one wallet to provide liquidity and others to capture it, potentially generating wash-trade volume for maker rebate farming.

4.3 Bot Cluster: Staggered Timing Offsets

Three wallets show identical behavior with staggered timing:

Table 11: Suspected single-operator bot cluster

Pseudonym	Trades	WR	Size	Avg Entry	Shared Mkts
Plaintive-Forgery	304	99.0%	10.00	278s	—
Nippy-Styling	141	98.6%	10.00	299s	57 w/ PF
Ambitious-Birch	132	100.0%	2.00	293s	81 w/ PF

All three wallets are 100% buy-only with exactly round trade sizes. Plaintive-Forgery enters at 278s, Ambitious-Birch at 293s, and Nippy-Styling at 299s—a **staggered cascade** where each bot fires 5–15 seconds after the previous one. They share 57–81 markets and trade the same side 100% of the time. This is almost certainly a **single operator** distributing trades across wallets with progressive timing to maximize fill probability while the direction is known.

5 Binance → Polymarket: The Information Channel

The Binance–Polymarket linkage is behavioral, not on-chain (Binance and Kalshi are off-chain KYC platforms with no public blockchain footprint—see Appendix B for on-chain tracing). Four signatures establish Binance as the information source:

1. **Timing conditionality:** Wallets trade *only* when Binance has moved >$80, achieving 97–100% WR vs. ∼50% on quiet windows.
2. **Sub-second entry:** Nippy-Styling at 299.2s average requires sub-second reaction to exchange feeds.
3. **Directional alignment:** Every latency bot trade matches the Binance-observed direction at trade time.

4. **Constant sizing:** Trade sizes are fixed regardless of conditions—automated execution triggered by exchange signals.

These wallets are not hedging on Binance (volumes too small). They use Binance as a **free information feed**.

6 15-Min Cross-Platform Analysis: Polymarket vs Kalshi

672 matched windows (664 with trades on both platforms). 2,536,367 Polymarket trades, 1,680,127 Kalshi trades.

6.1 Resolution Concordance: The 5.7% Problem

Table 12: Cross-platform resolution agreement

Metric	Value
Total matched windows	672
Both resolved	672
Agreement (same result)	**634 (94.3%)**
Disagreement (opposite result)	38 (5.7%)

38 windows disagree because Kalshi uses CF Benchmarks BRTI (60-second trailing weighted) while Polymarket uses Chainlink BTC/USD (point-in-time). When BTC oscillates near the start price, the oracles sample different values. This 5.7% rate means any hedged cross-platform position has a **1-in-18 chance of double loss**—"risk-free" arb is impossible.

6.2 Price Discrepancy Analysis

We compare average trade prices across platforms for each matched window:

Table 13: Price spread distribution (664 matched windows with trades on both platforms)

Spread Range	Windows	Percentage
$< 1\%$	25	3.8%
1–2%	16	2.4%
2–5%	52	7.8%
5–10%	107	16.1%
$> 10\%$	464	69.9%
Average spread	16.0%	
Maximum spread	48.3%	

The 16% average is inflated by price-reporting asymmetry (Kalshi reports YES-side only; Polymarket reports per-token execution prices, mixing UP and DOWN). Controlling for direction, the true spread is **1–3%**—within fee structure. Markets are already efficiently arbitraged.

6.3 Polymarket Wallet Profiling on Cross-Platform Windows

Of >20,000 Polymarket wallets active on matched windows: **1,631** have >60% win rate *and* >40% late-window concentration. 27 wallets achieve 100% WR with 30+ trades on these Kalshi-overlapping windows.

7 Discussion

7.1 Hypotheses: True vs. False

Table 14: Summary of hypothesis outcomes

H#	Hypothesis	Verdict	Evidence Strength
H1	Latency arb (exchange→PM)	**Confirmed**	Very strong ($p < 10^{-30}$)
H2	Sell-cluster arb	**Confirmed**	Strong (79% sell WR)
H3	Binance correlation	**Confirmed**	Very strong (97.9% vs 40%)
H4	Cross-platform signatures	**Confirmed**	Strong (6-wallet cluster)
H5	Market making	**Confirmed**	Very strong ($1.1M 2-sided)
H6	Divergence exploitation	**Confirmed**	Strong (93.5pp delta)

All six hypotheses confirmed. H1 + H3 together establish the mechanism: wallets watch Binance, wait for direction certainty, trade Polymarket in the final seconds.

7.2 Estimated Profits (5.6-Day Window)

- **Top 29 latency bots:** ∼$70K volume, ∼50% ROI/trade → ∼$35K profit. Annualized: $2–3M.
- **Market makers:** ∼$2.7M volume, 2–4% spread capture → ∼$54–108K.
- **Sell-cluster arb:** ∼$94K volume, 79% WR → ∼$15K profit.

8 Cross-Platform Trade Matching: Polymarket 15m ↔ Kalshi 15m

We match 672 Polymarket 15-min markets to 673 Kalshi KXBTC15M markets by start timestamp—**zero offset**, identical 15-minute schedules. Of these, 664 windows have trades on both platforms.

8.1 Price Divergence

Table 15: Price spread distribution: Polymarket vs. Kalshi 15-min (664 matched windows)

Spread Range	Windows	Percentage
$< 2\%$	30	4.5%
2–5%	82	12.3%
5–10%	206	31.0%
10–20%	194	29.2%
$> 20\%$	152	22.9%
Mean spread	6.94%	
Median spread	5.31%	
Maximum spread	**42.67%**	

The average 6.9% spread represents the mean absolute difference in YES-equivalent prices across platforms. In 52.1% of windows (346/664), the spread exceeds 5 cents—well above fee structure on both platforms. The maximum observed spread is 42.67% (Feb 23, 01:00 UTC: Polymarket YES at \$0.55, Kalshi YES at \$0.12).

Resolution concordance: Both platforms agree on the outcome in 94.3% of windows. The 5.7% disagreement rate (38 windows) arises from different settlement oracles (Chainlink vs. CF Benchmarks BRTI). Of the 664 windows with price data, 25.8% (171) were retrospectively arb-profitable—meaning a trader who took opposite positions based on the spread would have profited.

Table 16: Top 10 price divergences between Polymarket and Kalshi

Window (UTC)	P_YES	K_YES	Spread	P_Res	K_Res	Arb?
02-23 01:00	0.55	0.12	+0.427	down	no	YES
02-20 11:45	0.66	0.30	+0.362	down	no	YES
02-21 18:00	0.34	0.67	−0.331	down	yes	YES
02-21 19:30	0.68	0.35	+0.322	up	no	YES
02-25 11:15	0.92	0.61	+0.303	up	yes	—
02-19 12:00	0.77	0.48	+0.285	up	yes	—
02-21 00:45	0.49	0.23	+0.259	down	no	YES
02-22 09:15	0.42	0.67	−0.251	down	yes	YES
02-21 10:30	0.67	0.43	+0.244	up	no	YES
02-20 17:15	0.54	0.32	+0.221	down	no	YES

8.2 Second-Level Trade Synchronization

We match every Polymarket trade to every Kalshi trade within the same 15-minute window by timestamp, counting pairs that execute within 0 seconds (same second) and ±3 seconds.

Table 17: Cross-platform trade synchronization

Metric	Value
Total exact same-second pairs	**161,577**
Total within ±3s pairs	**216,338**
Most synchronized window	02-25 23:00 (89.2% of Poly trades matched)
Mean sync rate (top 20 windows)	78.3%

The top 10 most synchronized windows:

Table 18: Most synchronized cross-platform windows

Window (UTC)	Same-Sec Pairs	Near-Sync (±3s)	Sync %
02-25 23:00	356	389	**89.2%**
02-21 05:45	383	414	87.6%
02-25 01:45	377	422	87.3%
02-21 19:15	373	426	85.0%
02-21 02:00	374	413	82.4%
02-22 20:30	363	428	81.2%
02-25 00:15	382	433	75.8%
02-21 02:45	363	422	75.3%
02-23 01:30	382	424	72.9%
02-20 21:45	363	433	69.9%

In 20 windows, over 75% of all Polymarket trades have a Kalshi counterpart within 3 seconds. This density of same-second execution across two independent platforms is automated cross-platform trading.

8.3 Amount Matching: The Arbitrage Smoking Gun

We search for trade pairs across platforms where: (1) the dollar amount matches within ±20%, (2) the time difference is ≤5 seconds, and (3) the directions are **opposite** (buy UP on Polymarket while selling NO on Kalshi, or vice versa). Opposite-side, same-amount, near-simultaneous trades are the signature of cross-platform arbitrage.

Table 19: Amount-matched cross-platform trade pairs

Metric	Value
Total amount matches (±20%, ±5s)	1,581,658
Opposite-side matches (arb pattern)	**1,368**
Opposite-side ratio	68.7% of directional matches

Table 20: Opposite-side amount-matched trades: the arbitrage evidence

Window	P_USD	K_USD	Δt	P_Dir	K_Side	Poly Wallet
02-21 11:00	\$580.00	\$513.81	−4s	up	no	`0x7296...`
02-21 11:00	\$4.95	\$4.95	+2s	up	no	`0x03c3...`
02-20 13:45	\$101.25	\$92.07	−5s	up	no	`0x4586...`
02-20 13:45	\$49.65	\$49.50	−3s	up	no	`0x8e3d...`
02-20 16:45	\$4.95	\$4.95	+2s	up	no	`0x3d1b...`
02-20 16:45	\$4.95	\$4.95	+2s	up	no	`0xb373...`
02-21 08:00	\$49.38	\$53.46	−5s	up	no	`0x1da9...`
02-21 08:00	\$49.50	\$53.46	−5s	up	no	`0xf44d...`
02-20 04:45	\$3.00	\$3.17	0s	down	yes	`0xd95f...`
02-19 11:45	\$8.92	\$9.35	+5s	down	yes	`0x2518...`

Key patterns in the arb evidence:

- **Exact dollar matches:** Three instances of \$4.95/\$4.95 (identical amounts, 2 seconds apart, opposite sides on different platforms).
- **Large-size arb:** \$580 Polymarket UP buy paired with \$513 Kalshi NO trade, 4 seconds apart.
- **Same window, multiple wallets:** Feb 20 13:45 and Feb 21 08:00 each show 2+ wallets executing arb trades in the same window—different wallets racing the same signal.
- **Zero-second match:** \$3.00/\$3.17 trades execute in the **same second** on opposite sides of opposite platforms.
- **Wallet** `0x03c3b0236c...` appears in both amount matches and the core co-trading cluster (Table 22)—34,191 total trades, \$282K volume.

8.4 Intra-Platform Co-Trading: Wallet-Level Temporal Proof

Beyond cross-platform synchronization, we detect **wallet pairs on Polymarket that trade in the same second** with correlated direction. These are the actual wallets executing the arbitrage:

Table 21: Top same-second co-trading wallet pairs (Polymarket 15-min)

Wallet A	Wallet B	Same-Sec	Same Dir
`0xe041...f2e5`	`0x1d00...0313`	**11,869**	86.7%
`0x1979...9d`	`0x1d00...0313`	**11,306**	91.6%
`0x1979...9d`	`0xe041...f2e5`	**8,147**	84.9%
`0x03c3...1d`	`0xe041...f2e5`	6,349	80.2%
`0xe041...f2e5`	`0xbd7d...d1`	6,186	83.0%
`0xd0d6...93aa`	`0xe041...f2e5`	5,766	86.9%
`0x03c3...1d`	`0x1d00...0313`	5,746	73.1%
`0xd0d6...93aa`	`0x1d00...0313`	5,109	82.6%

`0xe041...f2e5` and `0x1d00...0313` execute **11,869 trades in the same second** with 86.7% directional agreement. `0x1979...9d` and `0x1d00...0313` execute 11,306 same-second trades with 91.6% same direction. These are not coincidences—they are bots reacting to the same signal (Binance BTC price) with sub-second latency.

The core cluster of 6 wallets executing in lockstep:

Table 22: Core arbitrage cluster: 6 wallets with coordinated temporal execution

Wallet	Trades	Volume	PnL	K_Align
0xe041d09715148a9a...f2e5	59,134	$353,079	$17,064	50.0%
0x1d0034134e339a30...0313	45,888	$1,252,304	$54,249	53.1%
0x03c3b0236c5a0105...1d	34,191	$282,217	$5,468	63.6%
0x1979ae6b7e6534de...9d	31,909	$1,544,952	$98,674	54.2%
0xd0d6053c3c37e727...93aa	20,902	$397,737	$30,307	50.0%
0xbd7d453564deca2b...d1	17,177	$839,424	$42,712	50.0%
Total	**209,201**	**$4,669,713**	**$248,474**	—

These 6 wallets account for **$4.7M in volume and $248K in profit** over 5.6 days, executing 209,201 trades with sub-second coordination. They share 30 common counterparties (market makers) across 8,000+ shared markets.

8.5 Behavioral Fingerprint Clusters

Three wallet pairs show near-identical timing fingerprints—average trade offset within 3–10ms and trade size ratios within 96–99%—indicating single-operator multi-wallet strategies:

Table 23: Behaviorally identical wallet pairs (likely same operator)

Wallet A	Wallet B	Offset Δ	Size Ratio
0x583a...ff6	0xabb4...ff4	3.4ms	95.9%
0xa07a...832	0x583a...ff6	9.4ms	99.1%
0x4396...32a	0x398f...264	1.2ms	86.7%

All of these wallets appear in the 100% Kalshi-aligned suspect list.

8.6 Wallet Performance with Kalshi Alignment

Each wallet's **Kalshi alignment score**: fraction of trades where bet direction matches the Kalshi outcome. Random = ∼50%.

Table 24: Top Polymarket wallets by PnL on 15-min markets

Wallet	Trades	WR	Volume	PnL	K_Align
0x1979...9d	31,909	54.3%	$1,544,952	$98,674	54.2%
0x1d00...13	45,888	53.1%	$1,252,304	$54,249	53.1%
0xbd7d...d1	17,177	50.1%	$839,424	$42,712	50.0%
0x63ce...9a	6,011	54.5%	$237,805	$34,113	54.3%
0x33fa...f4	10,577	77.5%	$187,575	$9,182	**81.9%**
0xc062...25	205	85.4%	$164,440	$6,913	**87.8%**
0x537...2e	2,422	69.0%	$44,141	$8,417	**71.1%**
0x03c3...1d	34,191	63.3%	$282,217	$5,468	**63.6%**

The top volume wallets show near-random Kalshi alignment (∼50–54%), consistent with market makers. However, a distinct cluster shows **anomalously high Kalshi alignment**:

Table 25: Wallets with 100% Kalshi alignment (20 wallets, all with 100% WR and 100% K_Align)

Wallet (full)	Trades	PnL
0x4602411d7a9ca3004813da8fec7b46d27d72b0f9	**900**	$303
0x9e51207b5fbb6cfe64d656ca73a4d8f2278b691c	**499**	$149
0xf2eda2eca20eea27bdb0038bc3fc559e2cbb98a8	282	$876
0x10e052124fdfe8a8b66811a4e0c4c19ffa26b38a	212	$216
0x70488a2e4635e21bff01fa084f8854e3d5e12fbc	173	$82
0x722beb62befdc30beafac5abae62d4683a6db7db	142	$64
0xa07a9fd88723a49e32761eef7b4677aaa3299832	109	$126
0x583a7e4e2ec324ccdb4811e66eb8168d9aa2aff6	93	$108
0x69786a14667cf010a87ce31ad4ea278b6a9bed38	85	$247
0xabb4affae8fef5eadaa3ec9ffd3e9bae18215ff4	64	$78
0x12eadd8029ffb1b53098d5bf439d9dc52db11774	56	$70
0xd26a07185f9075c541dc085e47c9bcbe9c8cc5ea	55	$77
0x398f2971ca851ef4560d172b2eee46be2eb38264	48	$109
0x0e60496876f522f15f370348c0fc22b70d68f1df	43	$46
0x0eb6d02bb5eb90f06d994fa8af7edfc6d9f01064	42	$51
0x4396378a91ba74be2b448f83c586574871fc332a	40	$191
0xd8fbd2007b6d7cd1ca19eceba1f1f28a64c1716a	39	$86
0x8830ce0cda490e27ecfcaf3c7817a45d3a452abf	59	$37
0x91176db78ea5d0d1566ee028120137e1bfaececa	33	$745
0xf3408ba8b1427564005fb5720c495f430958e55b	33	$636

20 wallets achieve 100% win rate with 100% Kalshi alignment. The most extreme: `0x4602...09` with **900 trades, every single one correct** ($p < 10^{-270}$). Total: **1,810 wallets** exceed 58% Kalshi alignment.

Note: wallets `0x583a...`, `0xa07a...`, `0xabb4...` from this 100%-aligned list are the *same wallets* that appear in the behavioral fingerprint clusters (Table 23), and wallets `0x4396...`, `0x398f...` form a 1.2ms-offset pair. These are the same operators running coordinated infrastructure across both Polymarket and Kalshi.

8.7 Temporal Arbitrage: Polymarket 5-min ↔ Binance

Complementing the Polymarket–Kalshi cross-platform analysis, we analyze Polymarket's 5-minute markets against real-time Binance BTC price data (929 windows):

Table 26: Binance-linked temporal arbitrage summary (5-min markets)

Metric	Value
Wallets analyzed	6,544
Arb suspects flagged	3,956
Wallets with >75% Binance alignment	1,247
Sync. volume bursts (Poly + Kalshi >2× avg)	72
Top wallet PnL	$86,800 (20,471 trades, 77.8% aligned)
Highest Binance-aligned WR	81.8% (1,132 trades)

72 windows show synchronized volume bursts where both Polymarket and Kalshi simultaneously exceed 2× their average volume—consistent with shared external triggers (large BTC moves on Binance) driving coordinated activity across both prediction markets.

9 Economic Analysis

9.1 P&L Decomposition by Strategy

The preceding sections identify distinct strategy types—latency arbitrage, sell-cluster minting, market making, and cross-platform hedging—but report only gross trade counts and win rates. Here we decompose realized profit and loss per strategy, accounting for the Polymarket fee structure, Polygon gas costs, and execution slippage. This analysis covers the 15-minute market dataset (2,536,367 trades, 20,620 wallets, 672 windows).

Fee model. Polymarket's NegRisk BTC markets operate with 0% maker fee (plus a small rebate), 0% taker fee, and no explicit resolution fee. The binding costs are:

1. **Spread cost:** the difference between execution price and fair value at trade time. For latency bots trading in the final seconds, fair value is near \$1 (direction known), so spread cost ≈ 0.
2. **Gas:** Polygon L2 transactions cost ∼\$0.01 each. At 2.5M trades, aggregate gas ≈ \$25,200.
3. **Slippage:** market impact from large orders. Relevant primarily for market makers with \$20M+ volume.

P&L computation. For each trade, realized P&L is:

$$\mathrm{PnL}_{\mathrm{buy}} = \begin{cases} s(1-p) & \text{if win} \\ -sp & \text{if loss} \end{cases} \tag{1}$$

$$\mathrm{PnL}_{\mathrm{sell}} = \begin{cases} sp & \text{if win} \\ -s(1-p) & \text{if loss} \end{cases} \tag{2}$$

where s is the number of shares and p is the execution price. We classify every wallet with ≥5 trades into one of five categories by its behavioral signature, then aggregate across all 2,521,731 classifiable trades.

Table 27: P&L decomposition by strategy (15-minute markets, Feb 19–25)

Strategy	Wallets	Trades	WR	Gross PnL	Volume	Gas	Net PnL	ROI
Latency Arb	494	54,299	93.5%	\$26,718	\$3,330,160	\$543	\$26,175	0.8%
Sell-Cluster	506	151,431	67.5%	\$28,549	\$1,096,530	\$1,514	\$27,035	2.5%
Market Making	3,255	1,831,650	52.4%	\$384,915	\$20,459,469	\$18,316	\$366,599	1.8%
Informed	1,180	230,991	87.1%	\$96,379	\$4,160,131	\$2,310	\$94,070	2.3%
Retail	7,956	253,360	43.2%	−\$211,975	\$4,420,859	\$2,534	−\$214,508	−4.9%
Total	13,391	2,521,731		\$324,587	\$33,467,150	\$25,217	\$299,369	

Per-trade economics. The key insight is that different strategies extract value through different mechanisms, with sharply different per-trade edge profiles:

Table 28: Per-trade economics by strategy

Strategy	Edge/Trade	Avg Size	Avg Price	Net/Trade
Latency Arb	\$0.492	\$73.00	0.92	\$0.482
Sell-Cluster	\$0.189	\$13.27	0.33	\$0.179
Market Making	\$0.210	\$27.36	0.58	\$0.200
Informed	\$0.417	\$26.54	0.85	\$0.407
Retail	−\$0.837	\$45.04	0.49	−\$0.847

- **Latency arb** (494 wallets, 93.5% WR): highest per-trade edge ($0.49) at an average entry price of 92¢—buying near certainty. These wallets trade $73 average size in the final minutes of each window. Despite the highest win rate, their total profit ($26K) is dwarfed by market makers because their capacity is constrained by available resting liquidity.
- **Sell-cluster** (506 wallets, 67.5% WR): moderate edge ($0.19/trade) via NegRisk minting at an average sell price of 33¢. Selling the loser token at p yields profit sp per winning trade but costs $s(1-p)$ per loss. At 67.5% WR, the strategy generates $28K net profit on $1.1M volume (2.5% ROI).
- **Market making** (3,255 wallets, 52.4% WR): lowest per-trade edge ($0.21) but highest aggregate volume ($20.5M) and profit ($367K net). Profit derives from the bid–ask spread across both UP and DOWN tokens. The 52.4% win rate implies a slight directional tilt beyond pure spread capture, consistent with informed market making.
- **Informed** (1,180 wallets, 87.1% WR): high win rate with moderate volume ($4.2M), generating $94K net. These wallets trade throughout the window (avg entry 579s, not concentrated in the final minutes) but show strong Kalshi outcome alignment, suggesting cross-platform information.
- **Retail** (7,956 wallets, 43.2% WR): negative expected value with −$212K aggregate P&L. The sub-50% win rate combined with the bid–ask spread produces −$0.85 per trade—the "house edge" of the prediction market.

9.2 Temporal Stationarity

A critical question for both traders and regulators: is the arbitrage opportunity stable, expanding, or being competed away? We decompose all metrics by calendar day.

Table 29: Day-by-day arbitrage activity (latency arb + sell-cluster + informed combined)

Date	Windows	Arb Trades	WR	PnL	PnL/Window	Active	New
Feb 19	96	68,102	82.5%	$21,681	$225.8	1,446	1,446
Feb 20	96	75,594	79.5%	$30,531	$318.0	1,491	211
Feb 21	96	53,660	77.2%	$27,505	$286.5	1,346	142
Feb 22	96	50,911	79.1%	$14,367	$149.7	1,394	133
Feb 23	96	64,993	82.1%	$19,475	$202.9	1,492	128
Feb 24	96	59,726	82.1%	$10,021	$104.4	1,511	70
Feb 25	96	63,735	84.4%	$28,067	$292.4	1,465	50

Stationarity assessment. We regress PnL-per-window on day index to test whether the opportunity is declining. The result: mean PnL/window = $225.66, slope = −$11.12/day—the slope is less than 5% of the mean, indicating a **stable** opportunity over the 7-day window. Daily PnL fluctuates between $10K and $31K, but this variance is driven by BTC volatility (more large moves = more arb opportunities), not by competition eroding the edge.

New entrant dynamics. 1,446 arb wallets appear on day 1, then the new-entrant rate declines sharply: 211 on Feb 20, falling to just 50 by Feb 25. Across the full period, 2,180 unique arb wallets enter. The declining entry rate suggests the niche saturated quickly: the first-mover operators dominate, and late entrants add little incremental volume. The active count stabilizes at ∼1,400–1,500 wallets per day by Feb 20—the market reaches a competitive steady state within 48 hours.

The opportunity is **structurally persistent** rather than transient, because the root cause—Binance BTC price leading Polymarket's Chainlink oracle by seconds—is a design feature, not an inefficiency that competition can eliminate. The only way to close the latency channel is for Polymarket to either reduce the settlement window or use a faster oracle.

9.3 Loss Distribution: Who Pays for Arbitrage?

Arbitrage profits are not created from nothing—they are extracted from counterparties. With $725K in gross arb wallet profits over 5.6 days (15-minute markets), we ask: who sits on the other side of these trades?

Caveat: single-venue visibility. We observe only one leg of each participant's strategy. Market makers on Polymarket almost certainly delta-hedge their binary option exposure on centralized venues (Binance perpetuals, Deribit options). A "loss" on Polymarket may be offset by a corresponding gain on the hedge leg, making the net position profitable through spread capture and maker rebates. The figures below therefore represent **adverse selection costs on Polymarket**, not necessarily economic losses. We use "loss" as shorthand for single-venue P&L throughout this section.

Methodology. For each of 672 market windows, we partition all 2.5M trades into five wallet categories. When an arb wallet wins, the counterparty is whoever bet on the losing side in the same window. We attribute losses proportionally: if market makers contributed 75% of losing-side volume and retail contributed 18%, the arb profit is attributed accordingly.

Table 30: Adverse selection attribution by counterparty type (15-minute markets, single-venue)

Counterparty	Arb-Attributed	Share	Losing Trades	Gross Loss	Loss Rate
Market makers	$542,627	74.8%	870,991	$6,334,443	2.65%
Retail/uninformed	$127,081	17.5%	151,819	$1,330,846	2.88%
Other arb bots	$55,303	7.6%	82,480	$573,365	—
Total	$725,011		1,105,290	$8,238,654	

Market makers absorb 74.8% of adverse selection. Of the $725K in arb profits, **$543K comes from market-maker counterparties**—the dominant source. This is the expected outcome in CLOB microstructure: market makers post passive limit orders on both sides; latency bots arrive with superior information and selectively take the stale side. The maker's resting order is filled at a price that no longer reflects fair value. This is identical to the adverse selection problem in traditional equity market making [1].

Critically, these single-venue, single-wallet figures likely overstate true MM losses because market makers can hedge *within the prediction market ecosystem itself*—where the payoff structures match perfectly (binary vs. binary, not binary vs. linear):

1. **Multi-wallet hedging on Polymarket.** NegRisk ensures 1 UP + 1 DOWN = 1 USDC. A market maker can mint both tokens for $1.00, then sell whichever side has demand at a markup, locking in the spread regardless of outcome. Alternatively, one wallet buys YES while another wallet (same operator) buys NO in the same window—our per-wallet P&L accounting attributes a "loss" to whichever wallet holds the losing token, but the operator's *combined* position may be flat or profitable. The six-wallet sell clusters identified in Section 4.3 (identical behavioral fingerprints, 24–27s timing σ) are likely exactly this: a single operator distributing both sides across wallets.
2. **Cross-platform hedging (Polymarket ↔ Kalshi).** Buy YES on Polymarket at $0.55, buy NO on Kalshi at $0.40—if the prices don't sum to $1.00, the position is profitable regardless of outcome. Section 7 documents exactly this activity: 1,368 opposite-side amount-matched trades across platforms. The 5.7% oracle disagreement rate is the residual risk, not a deal-breaker for MMs who can price it in.

Both hedges avoid the binary-vs-linear payoff mismatch that makes perpetual futures a poor hedge for short-window binary options (a perp's P&L scales with BTC move magnitude; a binary

settles 0 or 1 regardless of move size). The adverse selection cost we measure ($543K) is therefore an *upper bound* on true MM losses—the actual economic cost depends on how much of their book is internally hedged across wallets or cross-platform. Their real revenue comes from the bid-ask spread and Polymarket's 1–2% maker rebate on uninformed flow, which—even on a single-venue, single-wallet basis—already exceeds adverse selection costs ($385K net positive P&L on $20.5M volume).

The top 15 adverse-selection wallets are *all* classified as market makers, led by Square-Guy ($451K on-venue losses, 14,580 losing trades) and Canine-Commandment ($398K, 21,528 trades). On Polymarket alone, MMs remain net profitable ($385K aggregate P&L across $20.5M volume), meaning spread revenue from uninformed flow already exceeds adverse selection costs—even before accounting for off-venue hedging gains. Their effective adverse selection rate is 2.65¢ per dollar traded: **for every $100 in market-making volume, $2.65 is captured by arb bots**.

Retail traders: 17.5% of losses, 43.2% WR. Retail traders ($127K attributed to arb, $1.33M gross losses) bear a 2.88¢-per-dollar loss rate—slightly higher than market makers. With a 43.2% win rate against a 50% base rate, they lose −$0.85 per trade on average. Their $212K aggregate loss over 5.6 days ($4.4M volume) reflects the structural negative EV of uninformed participation: binary options with 50/50 base rates and positive transaction costs produce negative expected value for participants without an information edge.

Arb-on-arb losses: 7.6%. Even arb wallets lose $55K to other arb wallets—reflecting the competitive dynamics within the arb niche. When multiple bots chase the same signal, they compete on speed and the slower bot may get filled at a worse price or on the wrong side.

The value flow. Combining single-venue and cross-venue considerations, the economic flow is:

1. **Binance spot market** generates public BTC price information (free).
2. **Arb bots** ($151K combined profit from latency + sell-cluster + informed) convert that information into Polymarket profits by trading against stale limit orders.
3. **Market makers** ($543K single-venue, single-wallet adverse selection) are the primary counterparties. They recoup this through (a) bid-ask spread on uninformed flow, (b) Polymarket maker rebates (1–2%), and crucially (c) internal hedging across multiple wallets (NegRisk minting, both-sides quoting) and cross-platform hedging against Kalshi. Perp-based hedging is a poor fit due to the binary-vs-linear payoff mismatch on short windows. On Polymarket alone, single-wallet accounting already shows MMs net profitable ($385K)—true operator-level P&L is likely higher.
4. **Retail traders** ($127K attributed losses, 43.2% WR) face structural negative EV from the bid-ask spread and information asymmetry. Unlike MMs, retail participants are unlikely to hedge, making their losses closer to true economic losses.

This mirrors the latency arbitrage dynamic in traditional equities, where HFT firms exploit stale quotes on slower exchanges. The prediction market variant is more extreme because the binary outcome and fixed settlement window create a deterministic information advantage in the final minutes. However, the MM hedge channel means the on-chain "loss" figures represent an *upper bound* on true economic extraction—the real cost of the information asymmetry is ultimately borne primarily by retail participants and, to a lesser extent, by the hedging friction costs of market makers.

10 Out-of-Sample Validation (Feb 25–Mar 2)

To test whether the arbitrage patterns documented in Sections 3–8 persist beyond the initial observation window, we collected a second week of 15-minute cross-platform data (Feb 25–Mar 2, 2026). This validation period is fully out-of-sample: all thresholds, classifications, and hypotheses were fixed on week 1 data before week 2 collection began. The 5-minute Binance analysis was not re-run; those findings (Section 3–5) remain based on week 1 only.

10.1 Week 2 Dataset

Table 31: Week 2 dataset summary (Feb 25–Mar 2, 2026)

Source	Metric	Value
Polymarket 15-min	Trades	2,215,260
	Markets	576
	Unique wallets	19,138
Kalshi KXBTC15M	Trades	1,894,658
	Markets	633
Combined week 2 trades		4,109,918
Matched cross-platform windows		564

10.2 Cross-Platform Synchronization: Week 1 vs Week 2

The core finding—automated same-second trade execution across platforms—replicates cleanly:

Table 32: Cross-platform sync comparison

Metric	Week 1	Week 2
Matched windows	672	564
Same-second trade pairs	161,577	148,142
Within ±3s pairs	216,338	188,741
Top window sync rate	89.2%	89.1%
Oracle agreement	94.3%	93.8%
Mean price spread	6.94%	6.86%
Arb-profitable windows	25.8%	29.0%

Week 2 produces 148,142 same-second pairs (vs. 161,577 in week 1)—8.3% fewer, proportional to the smaller number of matched windows (564 vs. 672). Per-window sync intensity is essentially unchanged. The oracle disagreement rate ticks up slightly to 6.2% (vs. 5.7%), consistent with continued use of different settlement oracles (Chainlink vs. CF Benchmarks BRTI).

10.3 Tiered Amount Matching

Week 2 introduces a refined tiered matching methodology that relaxes dollar tolerance and time tolerance in stages, revealing the depth of cross-platform coordination:

Table 33: Tiered amount matching (week 2)

Tier	$ Tol.	Δt Tol.	Matches	Opp-Side	Opp%
Exact	±5%	±2s	245,292	125,682	51.2%
Tight	±10%	±5s	253,189	129,175	51.0%
Medium	±15%	±10s	139,267	71,381	51.3%
Loose	±20%	±15s	60,264	31,486	52.2%
Total			**698,012**	**357,724**	**51.2%**

The consistent ∼51% opposite-side rate across all tiers—slightly above the 50% random baseline—confirms that cross-platform hedging activity is real but represents a minority of the matched volume. The sheer scale (698K matches, 358K opposite-side) across just 564 windows demonstrates pervasive automated cross-platform execution.

10.4 Wallet Persistence

Do the same wallets operate across both weeks?

- **9,464 wallets** have 3+ amount-matched trades in week 2, led by `0xe041...f2e5` (25,393 matches)—the same wallet that topped the co-trading table in week 1 (Table 21).
- `0xabb4...5ff4` appears in the 100% Kalshi-aligned list in **both** weeks (64 trades in week 1, 68 in week 2), providing direct evidence of persistent cross-platform arbitrage infrastructure.
- Week 2 produces **20 wallets with 100% Kalshi alignment**, led by `0x2dce...305d` (2,052 trades, 99.6% WR, 100% K_Align). Total wallets with >58% alignment: 1,261 (vs. 1,810 in week 1).

10.5 PnL Stationarity: The Declining Edge

The most important validation finding: **arbitrage profitability is declining**.

Table 34: Day-by-day arbitrage activity (week 2, all arb strategies combined)

Date	Windows	Arb Trades	WR	PnL	PnL/Win	Active	New
Feb 25	96	61,707	85.0%	$27,589	$287.4	1,523	1,523
Feb 26	96	65,488	83.7%	$10,877	$113.3	1,603	201
Feb 27	96	62,264	78.7%	−$25,947	−$270.3	1,564	151
Feb 28	96	57,455	80.3%	−$6,736	−$70.2	1,515	93
Mar 1	96	69,207	75.4%	$11,274	$117.4	1,438	81
Mar 2	96	75,295	76.9%	$2,428	$25.3	1,477	50

Week 1 showed a stable opportunity: mean PnL/window = $225.66, slope = −$11.12/day (<5% of mean). Week 2 tells a different story: **mean PnL/window = $33.83, slope = −$31.37/day**. Two days (Feb 27, Feb 28) are outright negative. The latency arb strategy, which earned +$0.49/trade in week 1, drops to **−$0.31/trade** in week 2—the edge has inverted. Sell-cluster and informed strategies show similar degradation (Table 35).

Table 35: Per-trade economics comparison: week 1 vs week 2

Strategy	**W1 Edge/Trade**	**W2 Edge/Trade**	**W1 Net PnL**	**W2 Net PnL**
Latency Arb	\$0.492	−\$0.312	\$26,175	−\$12,541
Sell-Cluster	\$0.189	\$0.048	\$27,035	\$4,837
Market Making	\$0.210	\$0.200	\$366,599	\$300,573
Informed	\$0.417	\$0.113	\$94,070	\$23,276
Retail	−\$0.837	−\$0.690	−\$214,508	−\$159,808

Market making remains stable (∼\$0.20/trade in both weeks), consistent with structural spread capture rather than informational edge. Retail losses decrease in magnitude (−\$0.69 vs. −\$0.84/trade), possibly reflecting reduced participation as the market matures.

Interpretation. The structural mechanism persists (Binance price leads both platforms' oracles), but competition has eroded the extractable edge. The number of arb wallets stabilizes at ∼1,500/day—the same as week 1—but they split a shrinking pie. This is the expected lifecycle of a publicly visible arbitrage: structural persistence with economic decay.

11 Conclusion

We document two distinct arbitrage channels operating across cryptocurrency prediction markets, analyzing approximately 10.5 million trades over two weeks (Feb 19–Mar 2, 2026).

11.1 Channel 1: Binance → Polymarket (5-Minute Markets)

Binance BTC price serves as a **free information feed** for Polymarket's 5-minute binary options. We identify 1,647 latency arbitrage candidates, of which 29 achieve >98% win rates by trading in the final 15 seconds of each 300-second window. The evidence is unambiguous:

- Nippy-Styling trades at 299.2 seconds average (0.8s before resolution), 98.6% WR across 141 trades.
- Wallets achieve 97–100% WR when Binance has moved >$80, but ∼50% (random) on small moves.
- A staggered 3-wallet cascade (Plaintive-Forgery → Ambitious-Birch → Nippy-Styling) fires at 278s, 293s, 299s—likely a single operator.
- A coordinated market-maker (Capital-Potty, $94K volume) takes the opposite side of latency bots in 99.8% of shared markets.

Annualized, latency arbitrage on Polymarket's 5-minute BTC markets could yield $2–3M for a sophisticated operator.

11.2 Channel 2: Polymarket ↔ Kalshi (15-Minute Markets)

Second-by-second trade matching across 672 simultaneously active 15-minute windows reveals direct evidence of cross-platform arbitrage:

1. **161,577 same-second trade pairs** between Polymarket and Kalshi, with sync rates reaching 89.2% in individual windows.
2. **1,368 opposite-side amount-matched trades**—buy on one platform, sell on the other within ±5 seconds at matching dollar amounts (±20%). The $580/$513 pair (4 seconds apart, opposite sides) and exact $4.95/$4.95 pairs are textbook arb execution.
3. **1,810 Polymarket wallets with anomalous Kalshi alignment** (>58%), including wallets with 100% WR and 100% Kalshi alignment across hundreds of trades ($p < 10^{-270}$).
4. **Price divergences averaging 6.9%** (max 42.7%) between platforms, with 52.1% of windows exceeding 5 cents.
5. **72 synchronized volume bursts** where both platforms simultaneously exceed 2× average volume, triggered by Binance BTC moves.

However, cross-platform arbitrage is **not risk-free**: the 5.7% oracle disagreement rate (Chainlink vs. CF Benchmarks BRTI) means any hedged position faces a 1-in-18 chance of double loss. The residual alpha lies in the **speed-of-information contest**: monitoring Binance BTC prices in real-time and racing to trade on whichever prediction market offers the best effective price before resolution.

11.3 The Unified Picture

Both channels are driven by the same underlying dynamic: **Binance BTC price as the universal information source**. On 5-minute markets, traders watch Binance and race to Polymarket. On 15-minute markets, the same signal drives simultaneous activity on both Polymarket and Kalshi, creating the synchronized volume bursts and same-second trade pairs we observe.

The economic analysis (Section 8) decomposes $325K in net profits across five strategy types and shows the opportunity is temporally stable within week 1 (PnL/window slope <5% of mean). On-chain adverse selection analysis shows market makers are the primary counterparty (74.8%, $543K single-venue), but they likely offset most of this cost through spread capture, maker rebates, and delta hedges on centralized exchanges. Retail traders (17.5%, $127K, 2.88¢-per-dollar)

are the participants least likely to hedge, making them the ultimate bearers of the information asymmetry cost.

11.4 Two-Week Validation

The out-of-sample analysis (Section 9) confirms that the structural mechanism persists into week 2 (148,142 same-second pairs, 93.8% oracle agreement, 20 wallets with 100% Kalshi alignment) while revealing that the **economic opportunity is declining**: PnL/window drops from \$225.66 to \$33.83, with a trend of −\$31/window/day. Latency arbitrage—the highest per-trade edge in week 1—turns negative in week 2 (−\$0.31/trade). Only market making maintains stable returns across both periods. The arbitrage lifecycle is clear: structural persistence with economic decay as competition erodes the extractable surplus.

The wallets involved are listed in full in Appendix A. For on-chain identity tracing (Safe owner extraction, bridge scanning) of these wallets, see Appendix B.

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A Top Cross-Platform Arbitrage Suspect Wallets

Why no Kalshi wallets? Kalshi is a centralized, CFTC-regulated exchange. Its public trade API exposes `trade_id`, price, side, and contract count—but **no user identifiers**. Unlike Polymarket (where every trade is an on-chain transaction linked to a proxy wallet), Kalshi trades are anonymous by design. We therefore cannot identify the Kalshi-side operator of any cross-platform arbitrage pair. The wallets below are the **Polymarket side** of suspected cross-platform operations, identified by their anomalous alignment with Kalshi market outcomes and same-second trade synchronization.

#	Proxy Wallet	Trades	WR	K_Align	Pattern
15-min Perfect Kalshi Sync (100% alignment)					
1	`0x4602...b0f9`	900	100.0%	100.0%	$290K vol
2	`0x9e51...691c`	499	100.0%	100.0%	$6.6K vol
3	`0xf2ed...98a8`	282	99.7%	100.0%	$31K vol
4	`0x10e0...0f66`	212	100.0%	100.0%	$20.5K vol
5	`0x92b2...ef15`	209	100.0%	100.0%	$1.3K vol
6	`0x7048...8a42`	173	100.0%	100.0%	$3.9K vol
7	`0x722b...2de7`	142	100.0%	100.0%	$2.1K vol
8	`0xa07a...5b56`	109	100.0%	100.0%	$716 vol
9	`0x583a...e0e`	93	100.0%	100.0%	$620 vol
10	`0x6978...ed38`	85	100.0%	100.0%	$18.6K vol
15-min High-Volume Kalshi-Synchronized					
11	`0x03c3...a1d`	34,191	63.3%	63.6%	$282K vol, 5 matches
12	`0x33fa...97f4`	10,577	77.5%	81.9%	High vol + sync
13	`0xc062...ea25`	205	85.4%	87.8%	$164K vol
14	`0xf007...0b4`	7,988	70.0%	72.2%	Consistent sync
15	`0xa3d0...ec04`	3,334	67.0%	67.8%	$9.1K PnL
5-min Latency Arbitrage					
16	`0x571c...a61f`	20,471	—	77.8%	$86.8K PnL
17	`0x2b41...0d3`	1,132	—	81.8%	61.7% big-move WR
Top 15-min Volume					
18	`0x1979...9d`	31,909	54.3%	54.2%	$98.7K PnL
19	`0x1d00...0313`	45,888	53.1%	53.1%	$54.2K PnL

Kalshi-side limitation: The Kalshi operator behind these cross-platform pairs is *unidentifiable from public data.* The Polymarket-side evidence is strong: 10 wallets with 100% Kalshi outcome alignment across ≥85 trades each ($p < 10^{-25}$), plus 200 opposite-side amount-matched trade pairs with ≤5-second offsets. The same operator almost certainly controls accounts on both platforms, but linking them requires subpoena-level access to Kalshi's records.

Full results for all 7,570 wallets are in the companion file `data/cross_arb_15m_results.json`. The 1,810 wallets with >58% Kalshi alignment are listed with full trade statistics.

B On-Chain Wallet Tracing

All Polymarket wallets are Gnosis Safe proxy contracts on Polygon. We extracted Safe owners (signing EOAs) for 10 of 25 top arb proxies; the remaining 15 are MPC wallets with no on-chain owner. All 10 owners are unique—no shared EOAs across arb wallets. We scanned all owners across 7 bridge protocols (Wormhole, deBridge, Relay, LayerZero, Across, Socket, Etherscan multi-chain) and filtered 11 Polymarket infrastructure contracts (NegRiskCtfExchange, CTF Exchange, FeeModule, Relay Solver, GSN Relayer, etc.).

Proxy Wallet	Owner EOA	Activity
0xf2ed...98a8	0xd6c5...7d17	No multi-chain
0x4602...b0f9	0x8328...9847	Polygon only
0x9e51...691c	0x641c...7661	Polygon only
0x9117...eca	0x5dca...14da	Polygon only
0x1d00...0313	0xab6f...44d0	Polygon only
0x1979...9d	0x8055...f961	Polygon only
0x03c3...1d	0x508a...f707	Polygon + Arbitrum, **Mayan Swift**
0xd0d6...93aa	0x9698...d0e	No multi-chain
0x0ea5...17e4	0x66fb...6897	No multi-chain
0x2b41...0d3	0x46c8...eb26	Ethereum + Polygon, **Relay**

Key finding: Mayan Swift bridge links a Solana wallet to EOA `0x508a...f707` (owner of proxy `0x03c3...1d`, 34,191 trades). Bridge amounts total ∼$309 (gas funding only—not trading capital).

Result: 0 shared owners, 0 cross-wallet links, 0 Kalshi-linked addresses. Each arb wallet operates in on-chain isolation.

Full data in companion files:

- `data/proxy_owners.json` — Safe owner EOAs
- `data/multi_bridge_scan.json` — cross-chain bridge activity
- `data/solana_links.json` — Solana wallet links